Project Development Phase

Date	**06-03-2026**
Team ID	**Srasti Patel**
Project Name	**Heart Disease Analysis**

1. Data Preprocessing Steps

Before building dashboards in Tableau, the raw dataset was cleaned and transformed using **Tableau Prep Builder** to ensure accurate and meaningful analysis.

1.1 Data Import

- The heart disease dataset was imported into **Tableau Prep Builder**.
- Data sources included patient records containing demographic, medical, and diagnostic information.
- Initial schema and field types were reviewed to understand the structure of the dataset.

1.2 Handling Missing Values

- Identified columns with missing or null values.
- Missing values in important fields such as **age, cholesterol, blood pressure, and thalassemia** were handled by:
  - Removing records with excessive missing data.
  - Replacing null values with appropriate averages or default values where applicable.

1.3 Data Cleaning

- Removed duplicate patient records to maintain data integrity.
- Corrected inconsistent data entries and standardized categorical values.
- Ensured numeric fields such as **cholesterol, resting blood pressure, and heart rate** were correctly formatted.

1.4 Data Type Standardization

- Converted fields into appropriate formats:
  - Numeric fields → Integer/Float
  - Categorical fields → String
  - Binary indicators → Boolean or categorical

1.5 Feature Renaming and Labeling

- Renamed columns to more understandable business-friendly names.
- Example:
  - cp → Chest Pain Type
  - trestbps → Resting Blood Pressure
  - chol → Cholesterol Level

- thalach → Maximum Heart Rate Achieved

1.6 Data Transformation

- Created calculated fields for analysis:
  - **Heart Disease Risk Category**
  - **Age Group Segmentation**
  - **Cholesterol Risk Level**

1.7 Data Validation

- Verified aggregated values and distributions to ensure accuracy.
- Cross-checked cleaned data with the original dataset.

1.8 Data Export

- The processed dataset was published as a **clean dataset** and used as the input source for Tableau dashboards.

Business Questions with Corresponding Visualizations

1. Gender vs Heart Disease

Business Question:
Is heart disease more prevalent in males or females?

Visualization: Stacked Bar Chart (Sex vs Count of Heart Disease)

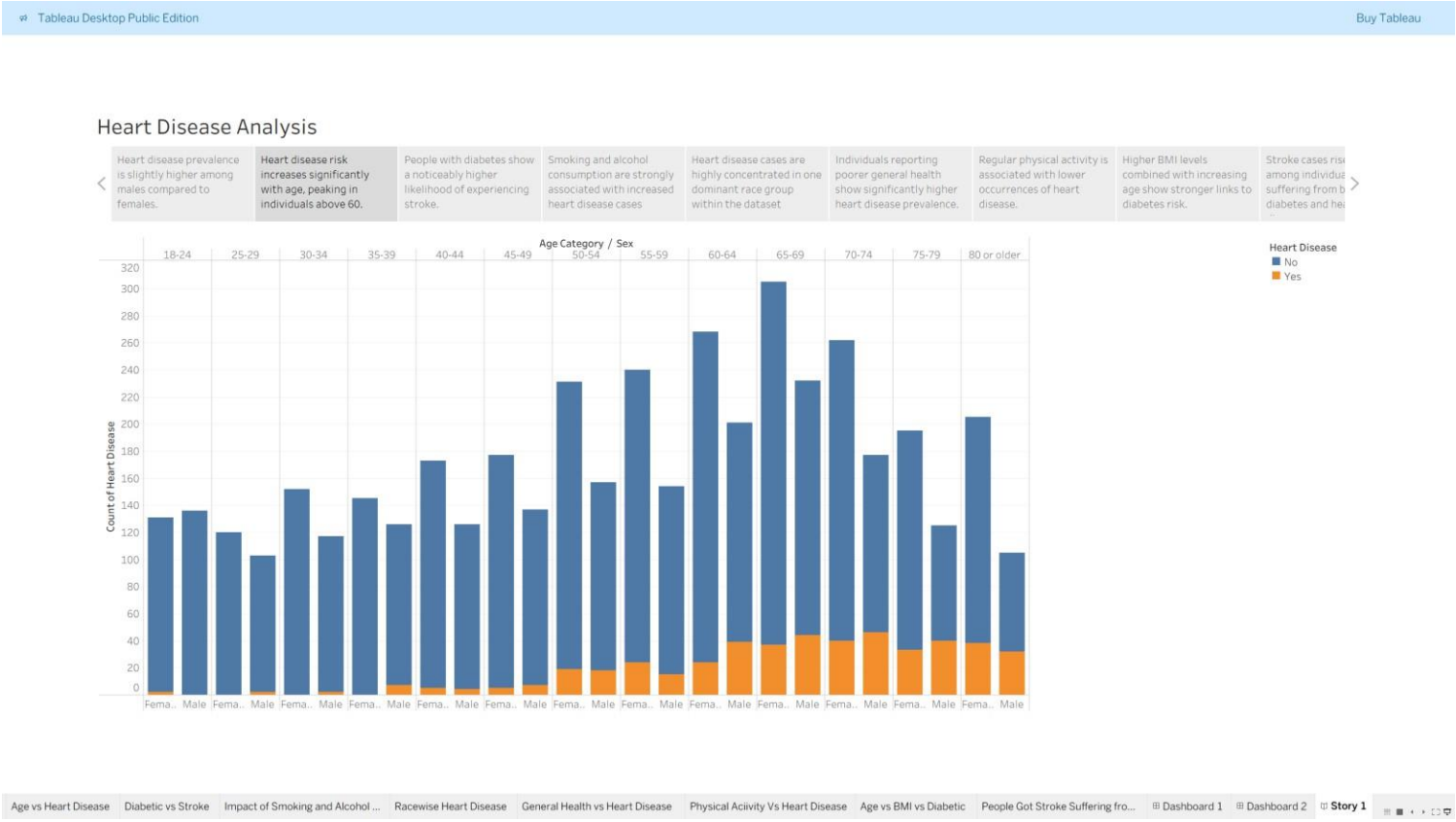
Purpose:
This visualization compares heart disease occurrence between male and female patients to identify gender-based risk patterns.

---

2. Age vs Heart Disease

Business Question:
How does heart disease risk vary across different age groups?

Visualization: Bar Chart (Age Category / Sex vs Count of Heart Disease)



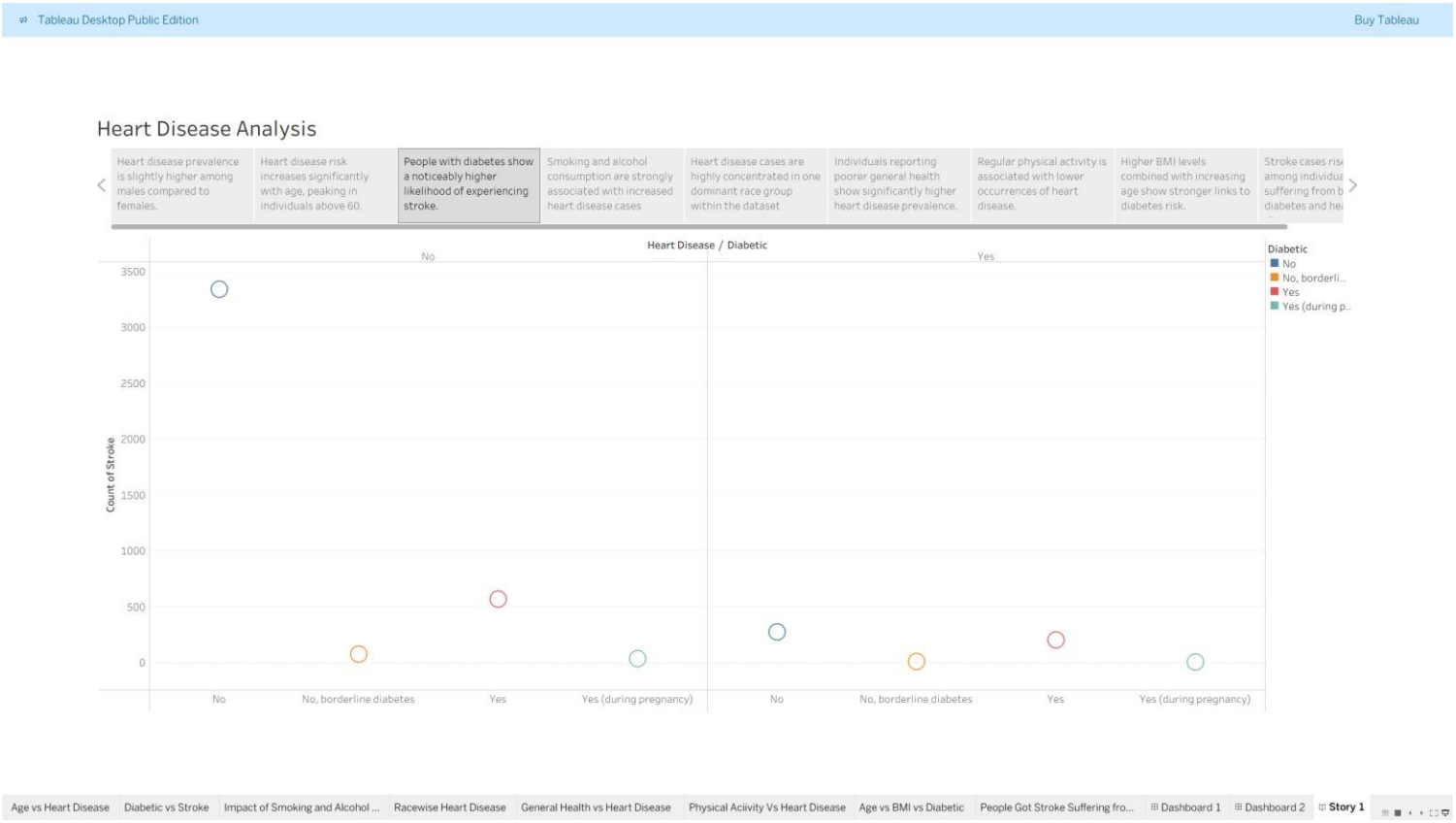
Purpose:
This chart shows how heart disease cases increase with age and highlights the age groups with the highest risk.

---

3. Diabetes vs Stroke

Business Question:
Are individuals with diabetes more likely to experience strokes and heart disease?

Visualization: Scatter / Bubble Plot (Heart Disease vs Diabetic vs Stroke)

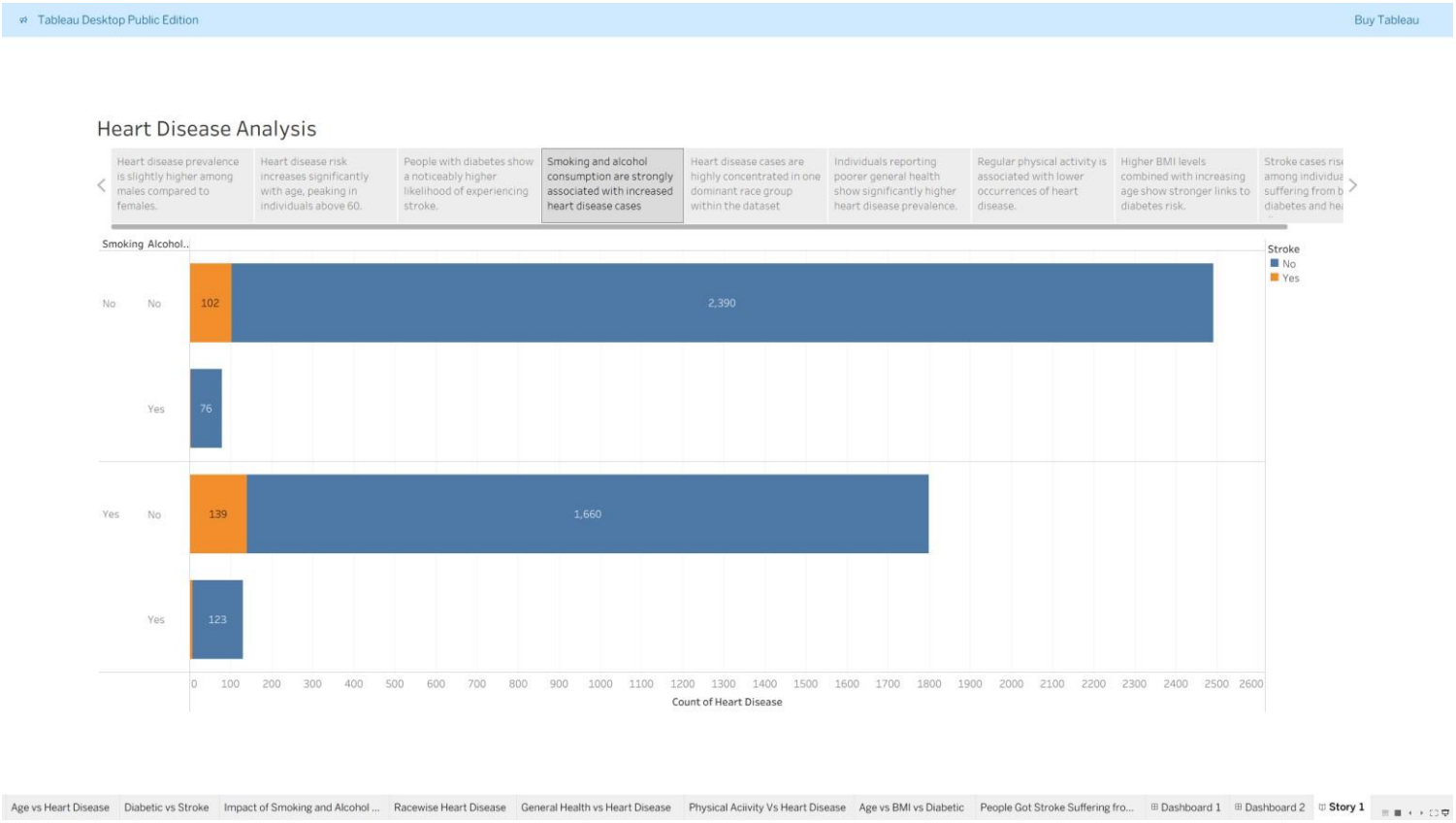


Purpose:
This visualization analyzes the relationship between diabetes status and the occurrence of stroke and heart disease.

4. Smoking and Alcohol Impact

Business Question:
Does smoking and alcohol consumption increase the likelihood of heart disease?

Visualization: Horizontal Stacked Bar Chart (Smoking/Alcohol vs Heart Disease Count)



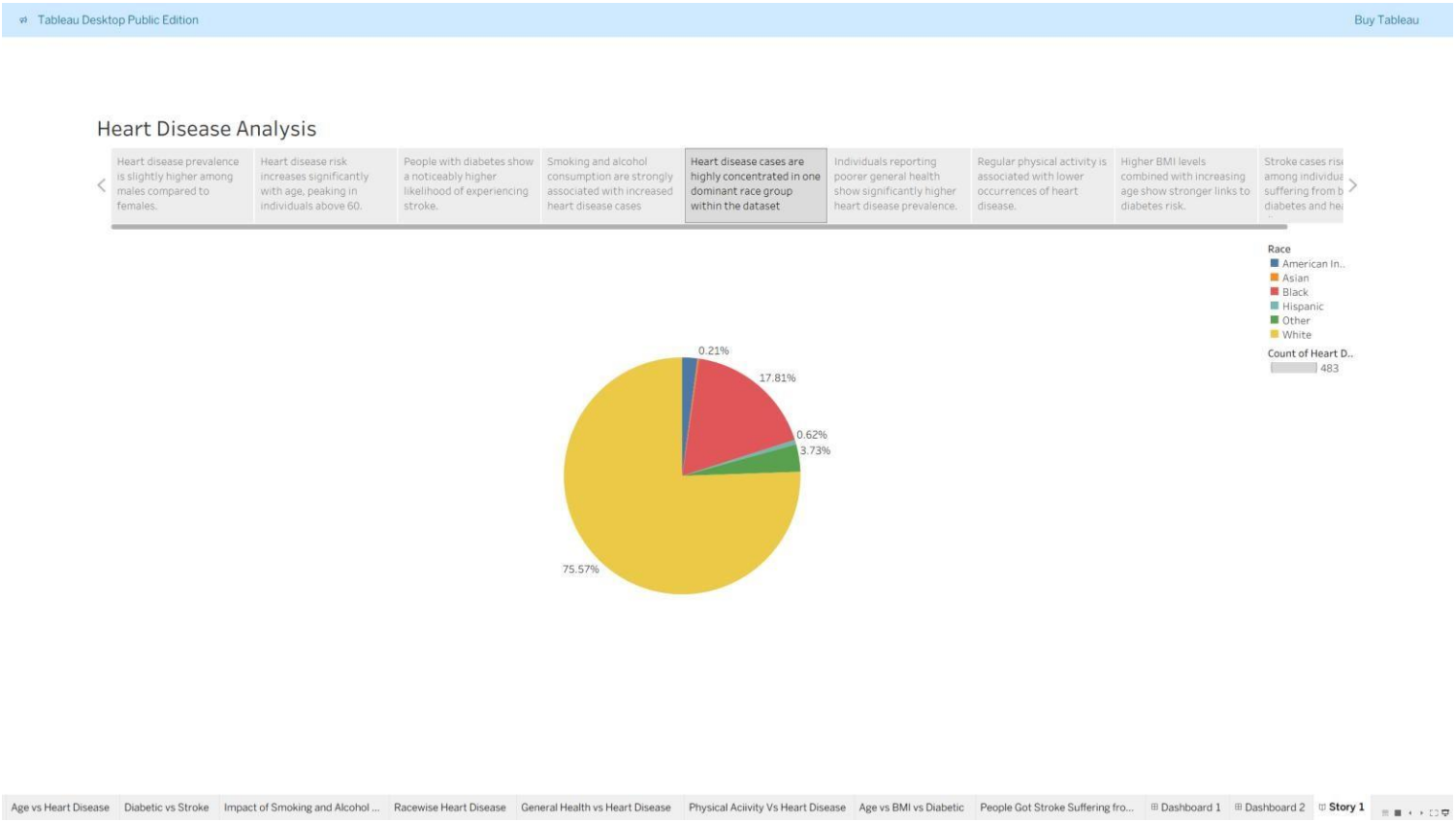
Purpose:
This chart compares heart disease cases among individuals who smoke or consume alcohol versus those who do not.

---

5. Race-wise Heart Disease Distribution

Business Question:
Which racial group shows the highest proportion of heart disease cases in the dataset?

Visualization: Pie Chart (Race vs Count of Heart Disease)



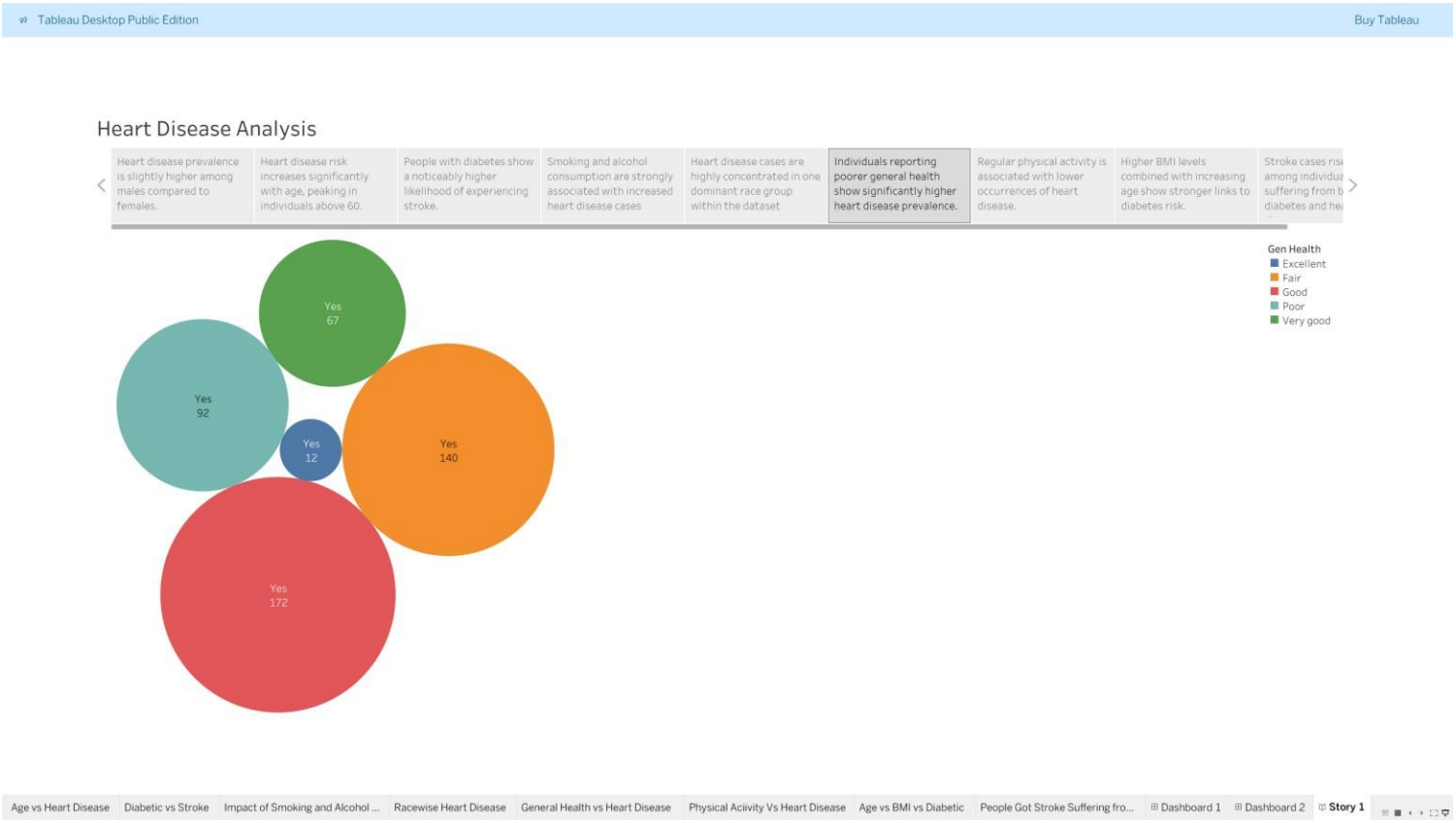
Purpose:
This visualization displays the percentage distribution of heart disease cases across different racial groups.

---

6. General Health vs Heart Disease

Business Question:
How does self-reported general health status influence heart disease prevalence?

Visualization: Bubble Chart (General Health Categories)



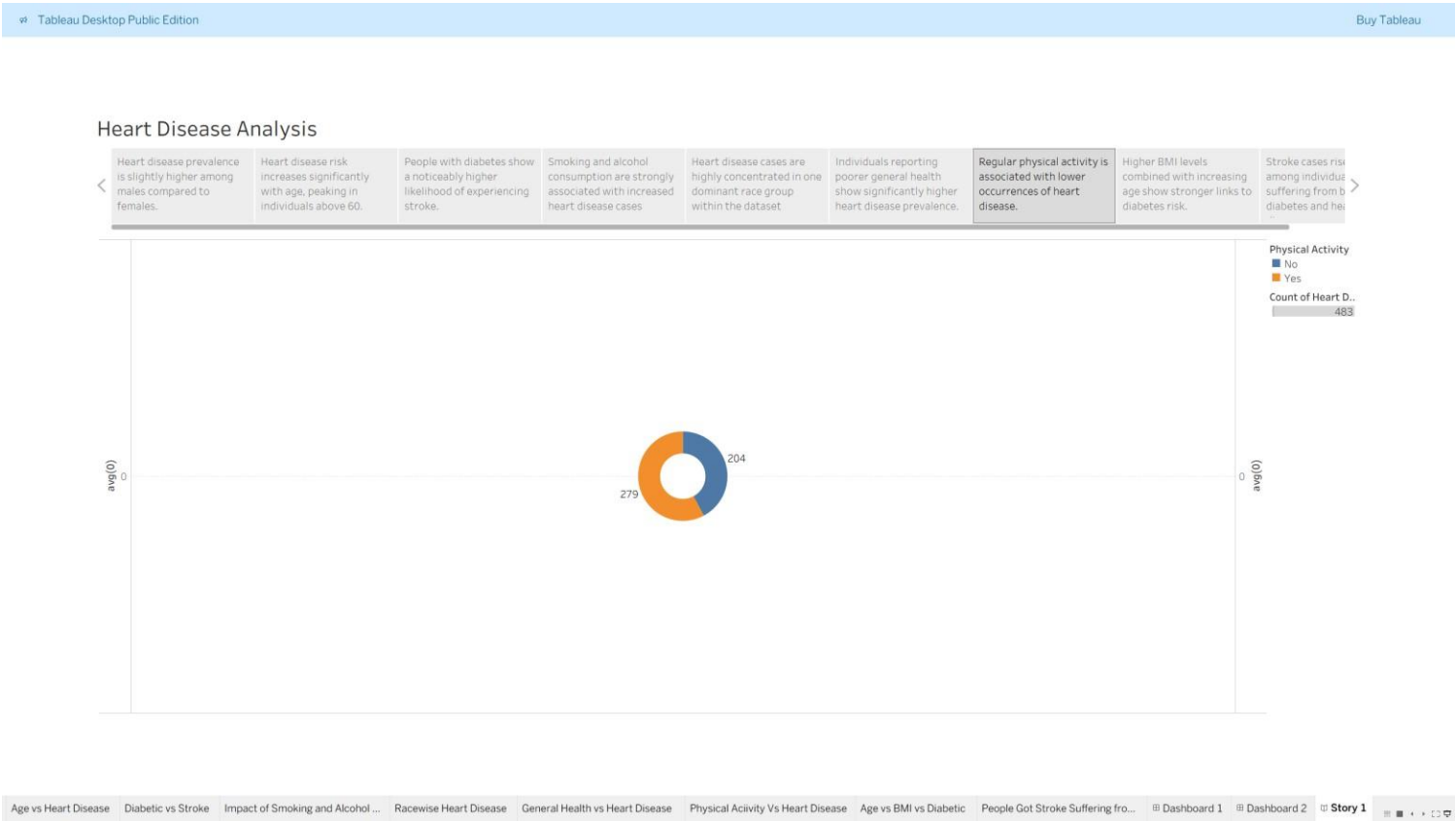
Purpose:
This chart illustrates how individuals reporting poorer health conditions tend to have higher heart disease occurrences.

---

7. Physical Activity vs Heart Disease

Business Question:
Does regular physical activity reduce the occurrence of heart disease?

Visualization: Donut Chart (Physical Activity vs Heart Disease Count)



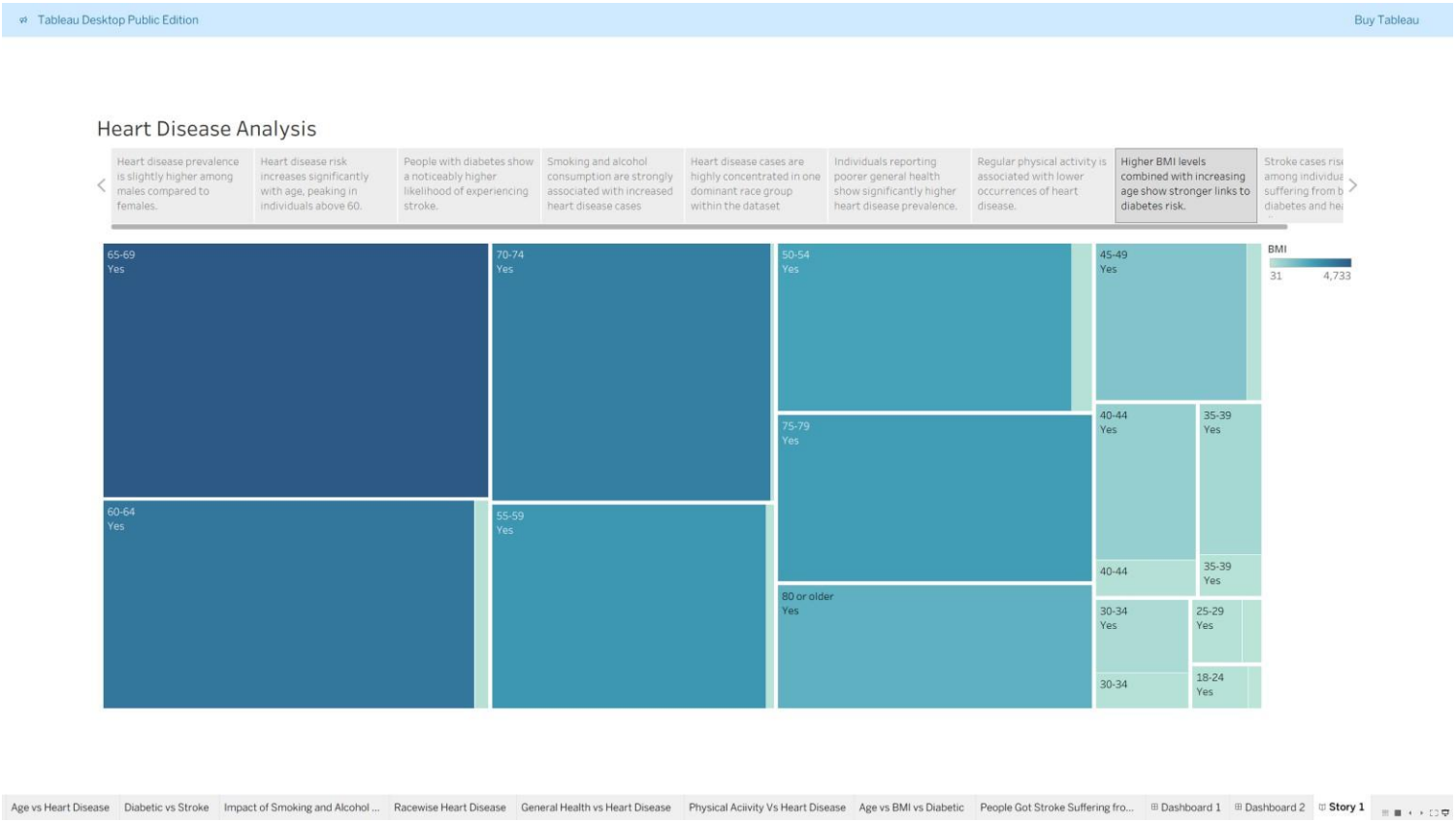
Purpose:
This visualization compares heart disease cases between physically active and inactive individuals.

---

8. BMI and Age vs Diabetes / Heart Disease

Business Question:
How do BMI levels combined with age affect heart disease and diabetes risk?

Visualization: Treemap (Age Groups vs BMI Levels)



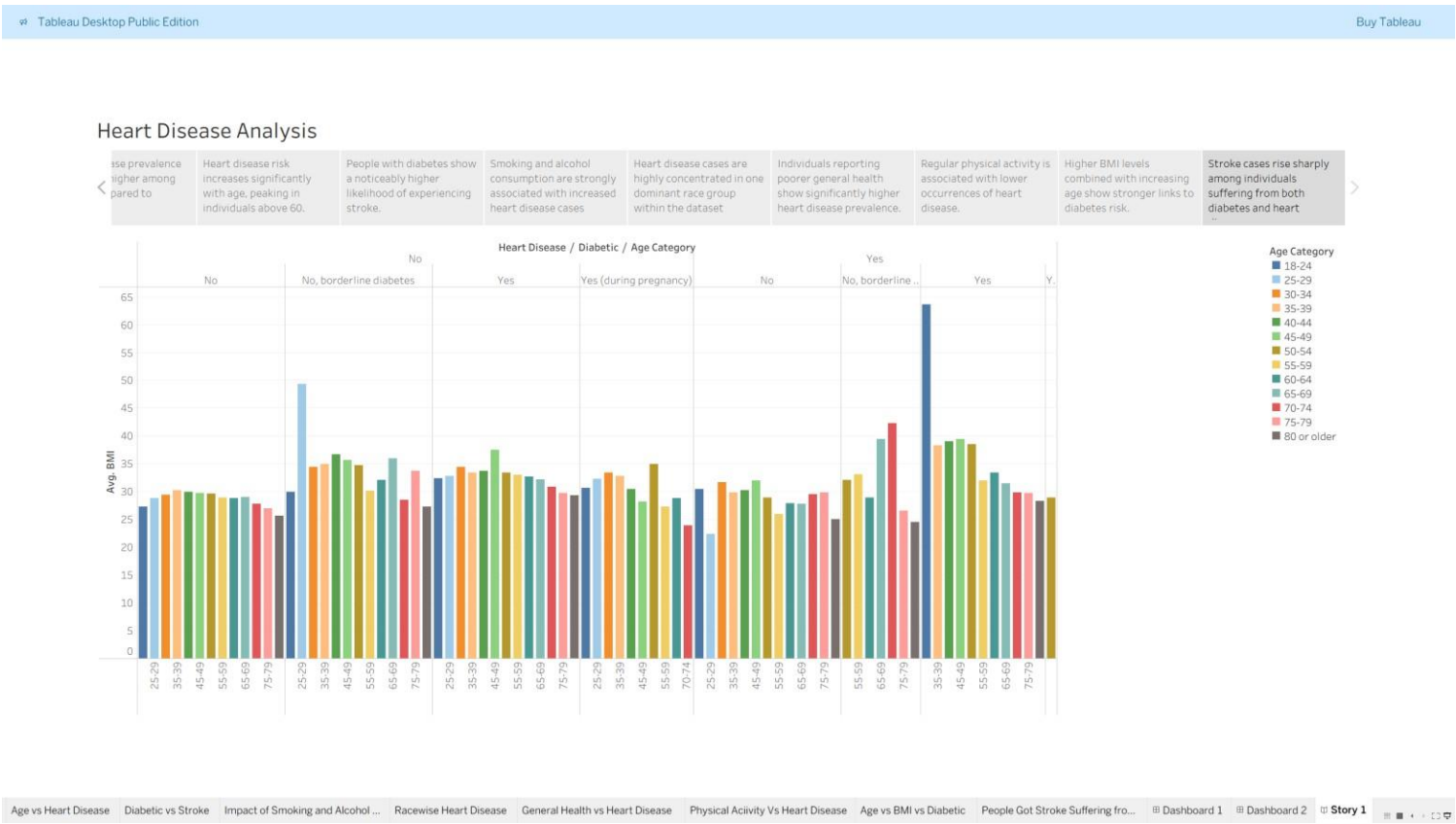
Purpose:
This treemap highlights which age groups with higher BMI levels show greater heart disease risk.

---

9. Age, BMI and Diabetes Combined Analysis

Business Question:
How does BMI vary across age groups and diabetic conditions in relation to heart disease?

Visualization: Multi-category Bar Chart (Heart Disease / Diabetic / Age Category vs Average BMI)



Purpose:
This visualization helps identify patterns where higher BMI levels correlate with diabetes and heart disease across different age groups.